

MIL-STD-29A

1 March 1962

SUPERSEDING
MIL-STD-29
23 September 1958

MILITARY STANDARD

**SPRINGS, MECHANICAL;
DRAWING REQUIREMENTS FOR**

APPENDIX

**OF REFERENCES TO
MATERIALS, DESIGN FORMULAS,
PROCESSES AND TOLERANCES
FOR MECHANICAL SPRINGS**

(APPENDIX INCLUDES NO MANDATORY PROVISIONS)

**UNITED STATES
GOVERNMENT PRINTING OFFICE
WASHINGTON: 1962**

MIL-STD-29A

**DEPARTMENT OF DEFENSE
DEFENSE SUPPLY AGENCY
WASHINGTON 25, D. C.**

**Springs, Mechanical; Drawing Requirements for
MIL-STD-29A**

1 March 1962

1. This standard has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy and the Air Force.

2. In accordance with established procedures, the Ordnance Corps, Bureau of Naval Weapons and the Air Force, have been designated as Army-Navy-Air Force custodians of this standard.

3. Recommended corrections, additions, or deletions should be addressed to the Standardization Division, Defense Supply Agency, Washington 25, D. C.

CONTENTS

1.	SCOPE
1.1	Scope
1.1.1	Types of Springs
1.1.1.1	Compression
1.1.1.2	Extension
1.1.1.3	Torsion
1.1.1.4	Flat
1.1.1.5	Constant Force
1.1.1.6	Garter
1.2	Purpose
2.	REFERENCED DOCUMENTS
2.1	Not Applicable
3.	DEFINITIONS
3.1	Mechanical Spring
3.2	Load
3.3	Total Deflection
3.4	Deflection per Coil
3.5	Set
3.6	Active Coils
3.7	Total Coils
3.8	Solid Length
3.9	Free Length
3.10	Pitch
3.11	Initial Tension
3.12	Spring Index
3.13	Spring Rate
3.14	Torque
3.15	Spring Tolerance
3.16	Factor of Safety
3.17	Direction of Helix
3.18	Reference Dimension
3.19	Stress Relieve
3.20	Range of Stress
3.21	Harden
3.22	Endurance Limit
3.23	Elastic Limit
3.24	Modulus of Rigidity

MIL-STD-29A

- 4. GENERAL REQUIREMENTS**
 - 4.1 Not Applicable
- 5. DETAIL REQUIREMENTS**
 - 5.1 General
 - 5.1.1 Delineation
 - 5.2 Specifications Applicable to All Springs
 - 5.2.1 Material Specifications
 - 5.2.2 Diameter of Wire
 - 5.2.3 Inspection Notes
 - 5.3 Format for Special Data Notes
 - 5.4 HELICAL COMPRESSION SPRINGS
 - 5.4.1 Definition
 - 5.4.2 Drawing Requirements
 - 5.4.2.1 Coil Diameter
 - 5.4.2.2 Direction of Helix
 - 5.4.2.3 Type of Ends
 - 5.4.2.4 Solid Length
 - 5.5 STRANDED WIRE COMPRESSION SPRINGS
 - 5.5.1 Definition
 - 5.5.2 Drawing Requirements
 - 5.6 HELICAL EXTENSION SPRINGS
 - 5.6.1 Definition
 - 5.6.2 Drawing Requirements
 - 5.6.2.1 Total Coils and Length Over Coils
 - 5.6.2.2 Maximum Extended Length
 - 5.6.2.3 Types of Ends
 - 5.6.2.4 Relative Position of Ends
 - 5.7 HELICAL TORSION SPRINGS
 - 5.7.1 Definition
 - 5.7.2 Drawing Requirements
 - 5.7.2.1 Total Coils and Length Over Coils
 - 5.7.2.2 Direction of Helix
 - 5.7.2.3 Pitch of Coils or Initial Tension
 - 5.7.2.4 Torque Loads
 - 5.7.2.5 Maximum Deflection without Permanent Set
 - 5.7.2.6 Spring Rate
 - 5.7.2.7 Type of Ends

MIL-STD-29A

- 5.8 SPIRAL TORSION SPRINGS
 - 5.8.1 Definition
 - 5.8.2 Drawing Requirements
 - 5.8.2.1 Outside and Inside Diameter of Coil
 - 5.8.2.2 Developed Length of Material and Active Length of Material
 - 5.8.2.3 Number of Coils in Free Position
 - 5.8.2.4 Type of Ends and Angular Relation of Ends
- 5.9 TORSION BAR SPRINGS
 - 5.9.1 Definition
 - 5.9.2 Drawing Requirements
 - 5.9.2.1 Diameter of Body and Ends
 - 5.9.2.2 Length Overall and of Ends
 - 5.9.2.3 Angle of Taper
 - 5.9.2.4 Torque Loads
 - 5.9.2.5 Spring Rate
 - 5.9.2.6 Type of Ends
 - 5.9.2.7 Direction of Windup Marking
 - 5.9.2.8 Angular Relation between Ends
- 5.10 VOLUTE SPRINGS
 - 5.10.1 Definition
 - 5.10.2 Drawing Requirements
 - 5.10.2.1 Outside and Inside Diameter
 - 5.10.2.2 Free Length
 - 5.10.2.3 Total Coils
 - 5.10.2.4 Radial Pitch
 - 5.10.2.5 Axial Pitch and Helix Angle
 - 5.10.2.6 Loads
 - 5.10.2.7 Solid Length
- 5.11 CONED DISC BELLEVILLE SPRINGS
 - 5.11.1 Definition
 - 5.11.2 Drawing Requirements
 - 5.11.2.1 Inside and Outside Diameters
 - 5.11.2.2 Free Height and Method of Stacking
 - 5.11.2.3 Loads
 - 5.11.2.4 Bearing Surfaces
 - 5.11.2.5 Bearing Surfaces Parallel
 - 5.11.2.6 Concentricity between Diameters
- 5.12 FLAT SPRINGS
 - 5.12.1 Definition
 - 5.12.2 Drawing Requirements
 - 5.12.2.1 Developed Lengths
 - 5.12.2.2 Clamp Length
 - 5.12.2.3 Loads

MIL-STD-29A

- 5.13 CONSTANT FORCE SPRINGS
 - 5.13.1 Definition
 - 5.13.2 Drawing Requirements
 - 5.13.2.1 Loads
 - 5.13.2.2 Initial Position and Final Position
 - 5.13.2.3 Type of Ends
 - 5.13.2.4 Fits Over Roller
 - 5.13.2.5 Thickness
- 5.14 GARTER SPRINGS
 - 5.14.1 Definition
 - 5.14.2 Drawing Requirements
 - 5.14.2.1 Shaft Diameter
 - 5.14.2.2 Compressive Load on Shaft

APPENDIX A

- 10. DESIGN OF MECHANICAL SPRINGS
 - 10.1 Purpose
 - 10.2 Scope

APPENDIX A SECTION I

- 11. SPRING MATERIALS
 - 11.1 General
 - 11.2 Sizes Available
- 12. HIGH CARBON STEELS
 - 12.1 General
 - 12.2 Music Wire
 - 12.3 Hard Drawn
 - 12.4 Oil Tempered
 - 12.5 Carbon Steel Valve Spring Quality
- 13. ALLOY STEEL WIRE
 - 13.1 General
 - 13.2 Chrome-Vanadium
 - 13.3 Silico-Manganese
 - 13.4 Chrome-Silicon
- 14. CORROSION RESISTING STEEL WIRE
 - 14.1 General
 - 14.2 Type FS302
 - 14.3 Type FS304
 - 14.4 Type FS316
 - 14.5 Type 17-7 PH
 - 14.6 Type FS420
 - 14.7 Type 431

MIL-STD-29A

- 15. COPPER BASE ALLOYS
 - 15.1 General
 - 15.2 Spring Brass
 - 15.3 Phosphor Bronze
 - 15.4 Beryllium Copper
- 16. NICKEL BASE ALLOYS
 - 16.1 General
 - 16.2 Monel
 - 16.3 K-Monel
 - 16.4 Inconel
 - 16.5 Inconel X
- 17. STEEL STRIP HIGH CARBON
 - 17.1 General
 - 17.2 SAE 1074
 - 17.3 SAE 1095

APPENDIX A SECTION II

- 20. SPRING DESIGN
 - 20.1 General
 - 20.1.1 Purpose
 - 20.1.2 Scope
 - 20.1.3 Abbreviations and Symbols
- 21. COMPRESSION SPRINGS
 - 21.1 Design Formulas
 - 21.2 Ends
 - 21.3 Diameter Changes
 - 21.4 Buckling
 - 21.5 Direction of Helix
 - 21.6 Natural Frequency, Vibration and Surge
 - 21.7 Impact
 - 21.8 Spring Nests
 - 21.9 Used As An Extension Spring
 - 21.10 Spring Index (D/d)
 - 21.11 Curvature Stress-Correction Factor (K)
 - 21.12 Keystone Effect
 - 21.13 Constants for Rectangular Wire
 - 21.14 Precautions and Suggestions for Effective Design
- 21.15 TABLES OF SPRING CHARACTERISTICS FOR COMPRESSION AND EXTENSION SPRINGS
 - 21.15.1 Example

MIL-STD-29A

- 21.16 DESIGN NOMOGRAPHS FOR COMPRESSION AND
EXTENSION SPRINGS
 - 21.16.1 Example
 - 21.17 Example of Compression Spring Calculation
 - 21.17.1 Necessity for Several Calculations
- 21.18 STRANDED WIRE HELICAL COMPRESSION SPRINGS
 - 21.18.1 General
 - 21.18.2 Stress
 - 21.18.3 Formulas for 3 Stranded Wire Spring
- 22. EXTENSION SPRINGS
 - 22.1 Deflection of Ends
 - 22.2 Stresses in Hooks
 - 22.3 Initial Tension
 - 22.4 Example of Calculation
 - 22.5 Precautions and Suggestions for effective Design
- 22.6 GARTER SPRINGS
 - 22.6.1 General
 - 22.6.2 Formulas
 - 22.6.3 Example of Calculation
- 23. TORSION SPRINGS
 - 23.1 Design Formulas
 - 23.2 Stresses in Ends
 - 23.3 Deflection of Ends
 - 23.4 Change in Diameter and Length
 - 23.5 Helix
- 23.6 DESIGN NOMOGRAPHS FOR HELICAL TORSION SPRINGS
- 23.7 MOMENT VS. WIRE SIZE CHART
- 23.8 Torsion Spring Calculation
 - 23.8.1 Example of Calculation
 - 23.9 Precautions and Suggestions for Effective Design
- 24. SPIRAL TORSION SPRINGS
 - 24.1 General
 - 24.2 Stress
 - 24.3 Ends
- 25. CONSTANT FORCE SPRINGS
 - 25.1 General
 - 25.2 Materials
 - 25.3 Inner Ends
 - 25.4 Outer Ends
 - 25.5 Multiple Mounting
 - 25.6 Stress Factor
 - 25.7 Extension Type

MIL-STD-29A

- CONSTANT FORCE SPRINGS — Continued
- 25.8 Formulas
- 25.9 Simplified Design
- 25.10 Design Charts
- 25.11 Example
- 26. FLAT SPRINGS
- 26.1 Design
- 26.2 Stress
- 27. CONED DISC (BELLEVILLE) SPRINGS
- 27.1 General
- 27.2 Load Deflection
- 27.3 Stress
- 27.4 Loads
- 28. RECOMMENDED MAXIMUM WORKING STRESSES
- 28.1 Fatigue Strength Curves
- 28.1.1 Light Service
- 28.1.2 Average Service
- 28.1.3 Severe Service
- 28.1.4 Other Materials
- 28.2 Permissible Elevated Temperatures

APPENDIX A SECTION III

- 30. SPRING MANUFACTURE
- 30.1 General
- 31. Stress Relieving
- 32. Cold Set to Solid
- 33. Grinding
- 34. Shot Peening
- 35. Protective Coatings
- 36. Hydrogen Embrittlement
- 37. TOLERANCES FOR SPRINGS
- 37.1 Tolerance on Loads

FIGURES

- Fig. DRAWING REQUIREMENT CHARTS AND DATA
- 1 Helical Compression Spring Forms
- 2(a) When No Load is Specified
- 2(b) When One Load is Specified
- 2(c) When Two Loads are Specified
- 2(d) When Spring Rate is Specified
- 3 Helical Compression Spring Ends
- 4(a) When No Load is Specified
- 4(b) When One Load is Specified
- 4(c) When Two Loads are Specified
- 4(d) When Spring Rate is Specified
- 5 Helical Extension Spring Forms

MIL-STD-29A

Fig.	DRAWING REQUIREMENT CHARTS AND DATA—<i>Cont'd</i>
6	Helical Extension Spring Ends
7	Helical Torsion Springs
8	Helical Torsion Spring Ends
9	Helical Torsion Spring Forms
10	Spiral Torsion Springs
11	Spiral Torsion Spring Ends
12	Torsion Bar Springs
13	Volute Springs
14	Coned Disc (Belleville) Springs
15	Method of Stacking Coned Disc (Belleville) Springs
16	Flat Springs
17	Constant Force Springs
18	Garter Springs

TYPICAL DETAIL DRAWINGS OF SPRINGS

19	Helical Compression
20	Stranded Wire Compression
21	Helical Extension
22	Helical Torsion
23	Spiral Torsion
24	Torsion Bar
25	Torsion Bar, Processing
26	Volute
27	Coned Disc (Belleville)
28	Flat
29	Constant Force
30	Helical Garter

APPENDIX A

31	Buckling of Helical Compression Springs
32	Surge Wave
33	Spring Nests
34	Stress Reduction in Spring Nests
35	Compression Spring Used As An Extension Spring
36	Constants for Rectangular Section

NOMOGRAPHS FOR HELICAL COMPRESSION AND EXTENSION SPRINGS

37	Fibre Stress VS. Load, Low Range
38	Fibre Stress VS. Load, High Range
39	Fibre Stress Correction for Curvature (K)
40	Deflection Per Coil VS. Load, Low Range
41	Deflection Per Coil VS. Load, High Range
42	Deflection Per Coil for Other Moduli
43	Stresses in Extension Spring Hooks

MIL-STD-29A**NOMOGRAPHS FOR HELICAL COMPRESSION AND
EXTENSION SPRINGS—Continued**

44	Stresses in Extension Spring Hooks Reduced
45	Permissible Torsion Stress Resulting From Initial Tension
46	Stresses in Torsion Spring Ends, Eye
47	Stresses in Torsion Spring Ends, Straight
48	Deflection of Torsion Spring Ends

NOMOGRAPHS FOR HELICAL TORSION SPRINGS

49	Fibre Stress VS. Moment, Low Range
50	Fibre Stress VS. Moment, High Range
51	Fibre Stress Correction for Curvature
52	Turns per Coil VS. Moment, Low Range
53	Turns per Coil VS. Moment, High Range
54	Turns per Coil for Other Moduli
55	Constant Force Spring Multiple Mounting
56	Constant Force Spring
57 to 60	Design Charts for Constant Force Springs
61	Coned Disc (Belleville) Springs, Constants
62	Allowable Working Stresses for Springs
63	Tolerances on Ends of Extension Springs

TABLES**APPENDIX A SECTION I MATERIALS****Table**

I	Preferred Sizes for Spring Materials
---	--------------------------------------

APPENDIX A SECTION II DESIGN

II	Formulas for Compression and Extension Springs
III	Compression Spring Formulas for Dimensional Characteristics
IV	Formulas for Load Deflections of Compression and Extension Springs (Impact)
V	Curvature Stress Correction Factors (K), for Compression and Extension Springs
VI	Tables of Spring Characteristics for Compression and Extension Springs
VII	Formulas for Torsion Springs
VIII	Moment VS. Wire Size Chart, (Torsion Springs)
IX	Formulas for Spiral Torsion Springs
X	Torque for 1 in. Wide Spiral Torsion Springs
XI	Stress Factor S _r
XII	Factor Q
XIII	Formulas for Flat Springs
XIV	Critical Stress Data
XV	Permissible Elevated Temperatures

MIL-STD-29A

APPENDIX A SECTION III MANUFACTURE

XVI	Recommended Tolerances on OD
XVII	Recommended Tolerances on Number of Coils
XVIII	Recommended Tolerances on Position of Arms of Torsion Springs
XIX	Recommended Tolerances on Length
XX	Recommended Tolerances on Squareness of Grinding for Compression Springs
XXI	Table of Third and Fourth Powers
XXII	Load Changes Due to Slight Wire Changes
XXIII	Causes of Spring Failure